

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Designed and built a resilient **Node.js/Puppeteer workflow engine** for MCA company-registration filings, turning browser-driven steps into repeatable backend operations.
- Integrated **public-site API lookups**, retries, session recovery, and operator checkpoints so partially completed workflows could resume instead of restarting, reducing manual filing time by **85%**.
- Implemented an **AI-assisted drafting flow** for company-name suggestions, object descriptions, and filing fields, with inspectable outputs and deterministic validation before execution.
- Containerized backend services with **Docker**, deployed them on **Google Cloud Run**, and automated releases with **GitHub Actions**, improving deployment velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Worked across InVesalious, an open-source **3D medical-imaging and neuronavigation application**, integrating FastSurfer CNN inference for **95 anatomical brain regions** with PyTorch/TorchScript and ONNX/TinyGrad paths.
- Handled **DICOM and volumetric MRI workflows** alongside NIfTI/MGZ data, including voxel conformation, orientation-sensitive preprocessing, plane-specific models, label remapping, LUT mapping, generated masks, and 3D surface inspection.
- Debugged **model-output orientation, label correctness, GPU/CPU paths, progress reporting, and 3D inspection** so segmentation results remained consistent across inference paths.
- Studied model architecture and runtime behavior while converting/integrating inference paths, then reduced processing time by **87%** through async/parallel execution with GPU/CPU, output-correctness, and QC checks.

Open Source

Contributions 

- Contributed maintainer-reviewed patches to **Invesalious** around medical-imaging artifact correctness and parser safety, with additional work in scientific data codebases.
- Also contributed to **Kuadrant MCP Gateway, Kyverno, Tiled, CRIU, IOOS, and VideoLAN**, covering backend failover behavior, security validation, data and parser correctness, and compatibility fixes.

Projects

Hyoka | Python, FastAPI, PostgreSQL, SQLAlchemy, OpenTelemetry, Next.js, Docker

GitHub 

- Built a framework-neutral **AI Reliability & Evaluation Platform** that records **tool/model/retrieval traces**, runs **evals and replays**, mines failures, and gates safer configurations before promotion.
- Implemented **OTLP/OpenInference-style ingestion**, an OpenAI-compatible proxy, evaluator registries, content-addressed artifacts, API keys, signed manifests, and **audit logs**.

Swish Support System | Python, Go, FastAPI, React, PostgreSQL, Redis, Langfuse

GitHub 

- Built a **policy-grounded AI support workflow** where LLM reasoning proposes actions, but deterministic services enforce refund, remake, coupon, trust, and escalation rules.
- Carried **customer/order/evidence context** across turns and traced high-risk decisions with Langfuse/local logs so agent behavior stayed inspectable instead of prompt-only.

Technical Skills

Languages: Python, TypeScript, Go, JavaScript

Frameworks: PyTorch, FastAPI, Next.js, React, SQLAlchemy, VTK, wxPython

Tools: ONNX, NumPy, PostgreSQL, Redis, Docker, OpenTelemetry, GitHub Actions